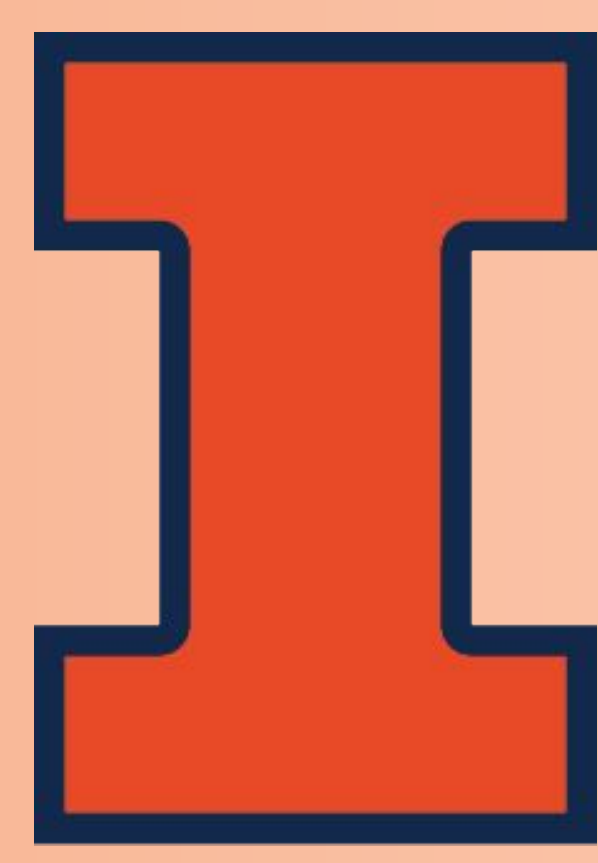




How Shoe Inserts Affect Lower Extremity Squat Mechanics

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1. Clinical motivation + open questions



Ankle sprains are one of the most common musculoskeletal injuries, especially in active populations. Up to **40% of ankle sprains** may lead to chronic symptoms such as pain, swelling, recurrent sprain, or instability [1]. A systematic review reported that **chronic ankle instability prevalence was about 46%** among individuals with a history of ankle sprain [2]. Recurrent ankle instability can limit **physical activity, sport participation, and daily mobility**. Conservative interventions such as **bracing, taping, and shoe inserts** are commonly used to improve support and reduce symptoms. However, the biomechanical effect of **orthopedic inserts** during functional movement is still not fully understood. Persistent ankle symptoms can affect return to sport, exercise participation, and overall quality of life. Understanding how inserts affect movement may help guide conservative support strategies for individuals with ankle instability or altered foot mechanics.



- **Ankle sprain / instability**
- **Chronic symptoms or reduced activity**
- **Use of external support devices**
- **Understand biomechanical effects of inserts**

Open question: Can orthopedic inserts change ankle, knee, and hip biomechanics during squatting?

2. Technical background

Squatting requires coordinated control of the **ankle, knee, and hip** to maintain balance and support body weight. Lower-extremity squat mechanics are influenced by **muscular strength, joint flexibility, and neuromuscular control** rather than body size alone [3]. The **foot and ankle** are the main mechanical interface between **the body** and the ground. **Orthopedic inserts** may modify plantar pressure, foot alignment, and force distribution during weight-bearing tasks [4]. Prior studies suggest that orthoses, taping, and insoles can influence **postural stability** and lower-extremity biomechanics [5], [6].

3. Background for objective

Changes in **underfoot support** may affect ankle mechanics during squatting. Altered ankle mechanics may also influence **knee and hip joint movement strategies**. **Joint angles** describe movement patterns, while **joint torques** estimate joint loading. Comparing squats with and without inserts helps determine whether inserts meaningfully change lower-extremity biomechanics.

Objective

Evaluate the effect of orthopedic inserts on ankle, knee, and hip joint angles and torques during a squat.

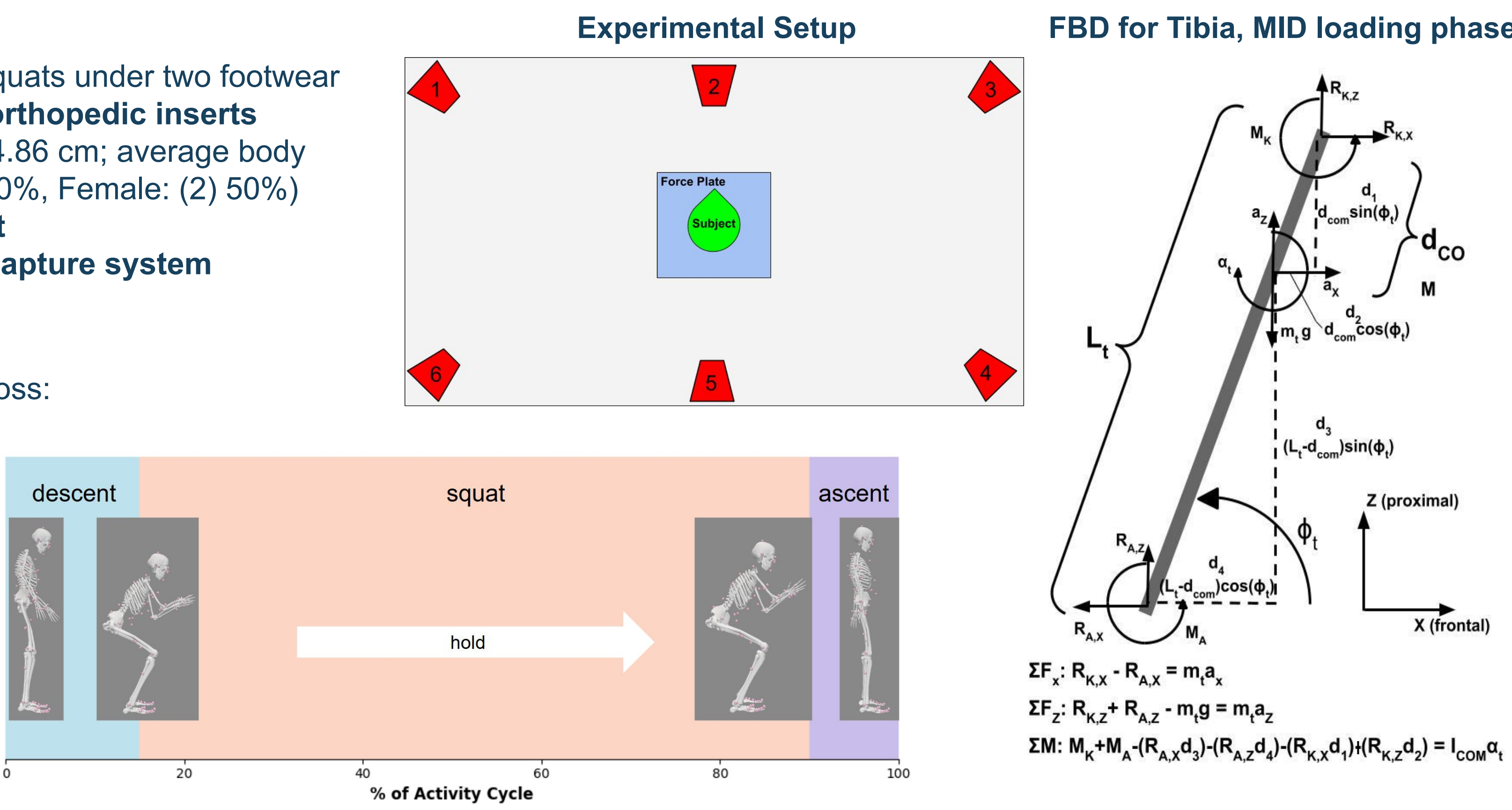
4. Methods

Experiment setup
n = 4 healthy adults completed bodyweight squats under two footwear conditions: **flat shoes only** and **flat shoes + orthopedic inserts** (average age = 22 years; average height = 164.86 cm; average body mass = 67.38 kg; sex distribution = Male: (2) 50%, Female: (2) 50%)
3 trials per condition; 6 trials per participant
Motion recorded using a **six-camera motion capture system**

Sagittal plane
Squat motion analyzed in the **sagittal plane**
Hip, knee, and ankle mechanics evaluated across:
descent → hold → ascent

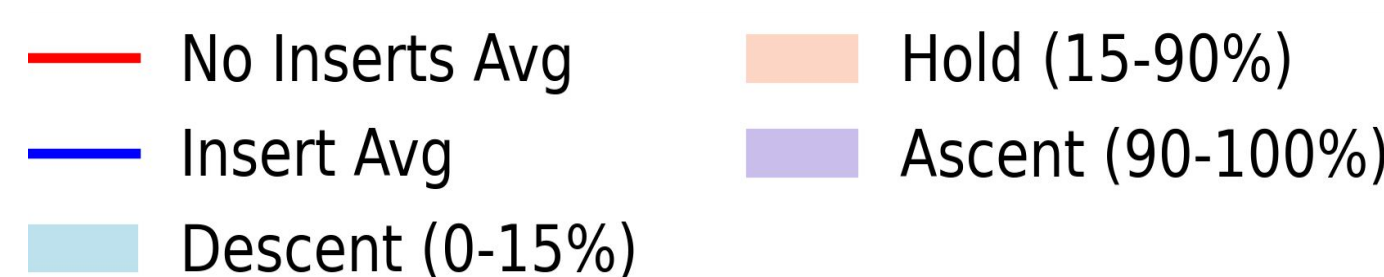
Variables measured

Variable	Measurement
Joint angles	Hip, knee, ankle
Joint torques	Hip, knee, ankle
Condition	Flat shoes vs. inserts
Phase	Descent, hold, ascent

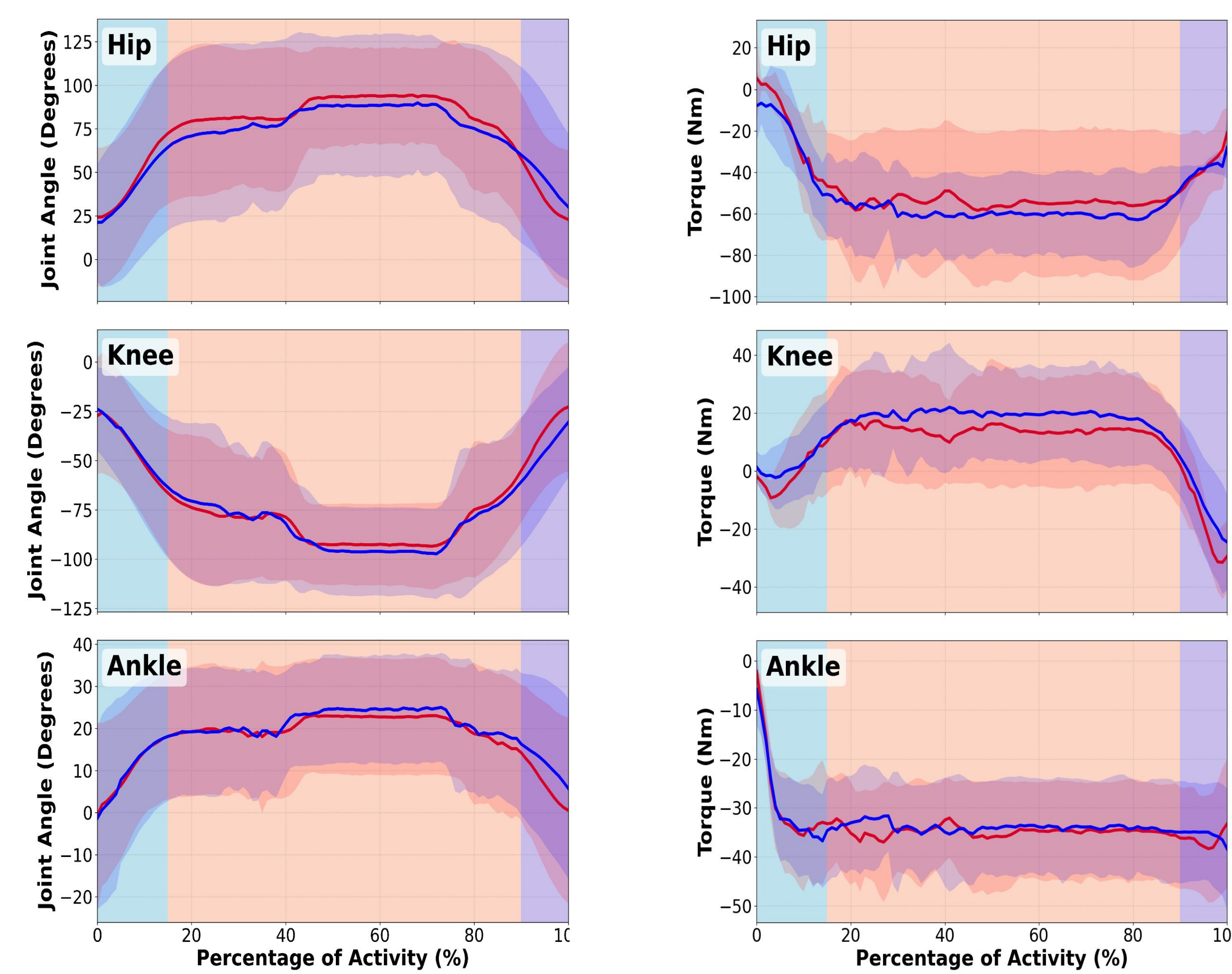


5. Key findings

No observed significant difference in joint angles throughout activity. The insert condition observed slight differences in joint torque at knee and hips at the hold phase. **Between 40% and 80% activity**, joint angles flatten out.



Bilateral Average Joint Angles with ±1 SD Bilateral Average Joint Torques with ±1 SD



6. Conclusions

We expected the inserts to directly influence ankle mechanics, however **no significant differences** were observed. **During the hold phase**, a slight variation in **knee and hip joint torque** was observed. Therefore, further exploration of torque and power should be studied. This could be due to an altered force distribution caused by the inserts, leading to different muscles activations
Future studies: Explore muscle activation (EMG) to understand the different muscles activated in each condition. A larger sample size will help with improving the accuracy and decrease the variability in between participants.
Limitations: Our project was centered around ankle stability, having more accurate motion capture data of the ankle would benefit this analysis. Current data collection methods generated stochastic motions in OpenSim which may have affected the resulting force and angle data.

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